2026-08-15

SINGLE-PARCEL HEAT DUE DILIGENCE

Site: my_parcel_indianapolis (-)

Lot: 75.38 acres · 655 x 712 m

AOI: 3.21 km2 · 891 tiles at 60 m

Parcel peak: 31.54 °C

FORTYGUARD

EXECUTIVE SUMMARY

my_parcel_indianapolis (-) — , . 75.38-acre parcel, zoning n/a, proposed use n/a.

Metric	Value	Reference
Parcel daily peak (area-weighted)	31.54 °C	no city baseline
Parcel daily mean	25.76 °C	AOI mean 25.78 °C
Diurnal swing	9.50 °C	peak minus daily minimum
Hours above 27 °C	72.0 h of 168 h	43% of window
Longest consecutive run	7.0 h	overnight recovery at <12 h
Impervious share (vicinity)	96.9%	EPA trigger >60%
Tree canopy (vicinity)	3.1%	USDA trigger <15%
Heat index at hot hour (18:00)	45.6 °C	NOAA Extreme Caution 32 °C
Peak apparent temperature	33.4 °C	at 18:00
Peak wet-bulb	25.5 °C	evaporative limit 26 °C

Method. A single heatmap was run over the parcel plus a 500 m context ring (3.21 km², 891 tiles at 60 m). Parcel figures are the **area-weighted mean** of the 108 tiles the boundary overlaps, weighted by overlap area — not a centroid lookup. At 60 m the parcel spans roughly 84.7 tiles, so a nearest-tile reading would discard most of the site.

Why duration leads. Across this AOI the daily-peak snapshot varies by only 0.01 °C, while hours-above-threshold varies by 0.0 h. At parcel scale the duration layers, not the snapshot, carry the signal.

FINDINGS AND RECOMMENDED ACTIONS

Each row is a published threshold, the measured value against it, and the program the action comes from. No composite index is used.

Trigger	Measured	Threshold	Verdict
Hours above 27 °C over 168 h	72.0 h (43% of window)	27 °C (NOAA Caution)	ELEVATED
Longest consecutive run above threshold	7.0 h	12 h (overnight recovery)	RECOVERS OVERNIGHT
Impervious surface share (site vicinity)	96.9%	60%	EXCEEDS
Tree canopy share (site vicinity)	3.1%	15%	BELOW
Ground-level vegetation at frontage	15.4% of field of view	10% (pedestrian shade)	ADEQUATE
Heat index at the hot hour (18:00)	45.6 °C	32 °C (Extreme Caution)	EXCEEDS
Construction-phase worker exposure at peak hour	45.6 °C at 18:00	32.2 °C (OSHA high-heat)	TRIGGERED
Peak wet-bulb temperature	25.5 °C	26 °C (evaporative-cooling limit)	BELOW

Hours above 27 °C over 168 h — ELEVATED

Cooling load is duration-driven, not peak-driven; model annual hours, and specify envelope for sustained rather than instantaneous load.

Program: ASHRAE 90.1 / ASHRAE 55 — Thermal Environmental Conditions

Longest consecutive run above threshold — RECOVERS OVERNIGHT

Exposed thermal mass remains a viable passive strategy on this site.

Program: EPA Heat Island Reduction — Cool Roofs

Impervious surface share (site vicinity) — EXCEEDS

Specify high-albedo roofing and cool or permeable paving at design stage.

Program: EPA Heat Island Reduction — Cool Roofs & Cool Pavements

Tree canopy share (site vicinity) — BELOW

Include a planting plan in the landscape package; quantify the canopy benefit before entitlement hearings.

Program: USDA Forest Service — i-Tree

Ground-level vegetation at frontage — ADEQUATE

Frontage already offers usable pedestrian shade.

Program: NACTO Urban Street Design Guide

Heat index at the hot hour (18:00) — EXCEEDS

Occupant-comfort risk at peak; specify to ASHRAE 55 and plan shaded outdoor amenity.

Program: ASHRAE 55 — Thermal Environmental Conditions

Construction-phase worker exposure at peak hour — TRIGGERED

Build a heat-illness prevention plan into the construction schedule and bid documents — mandatory rest, shade, and water provisions.

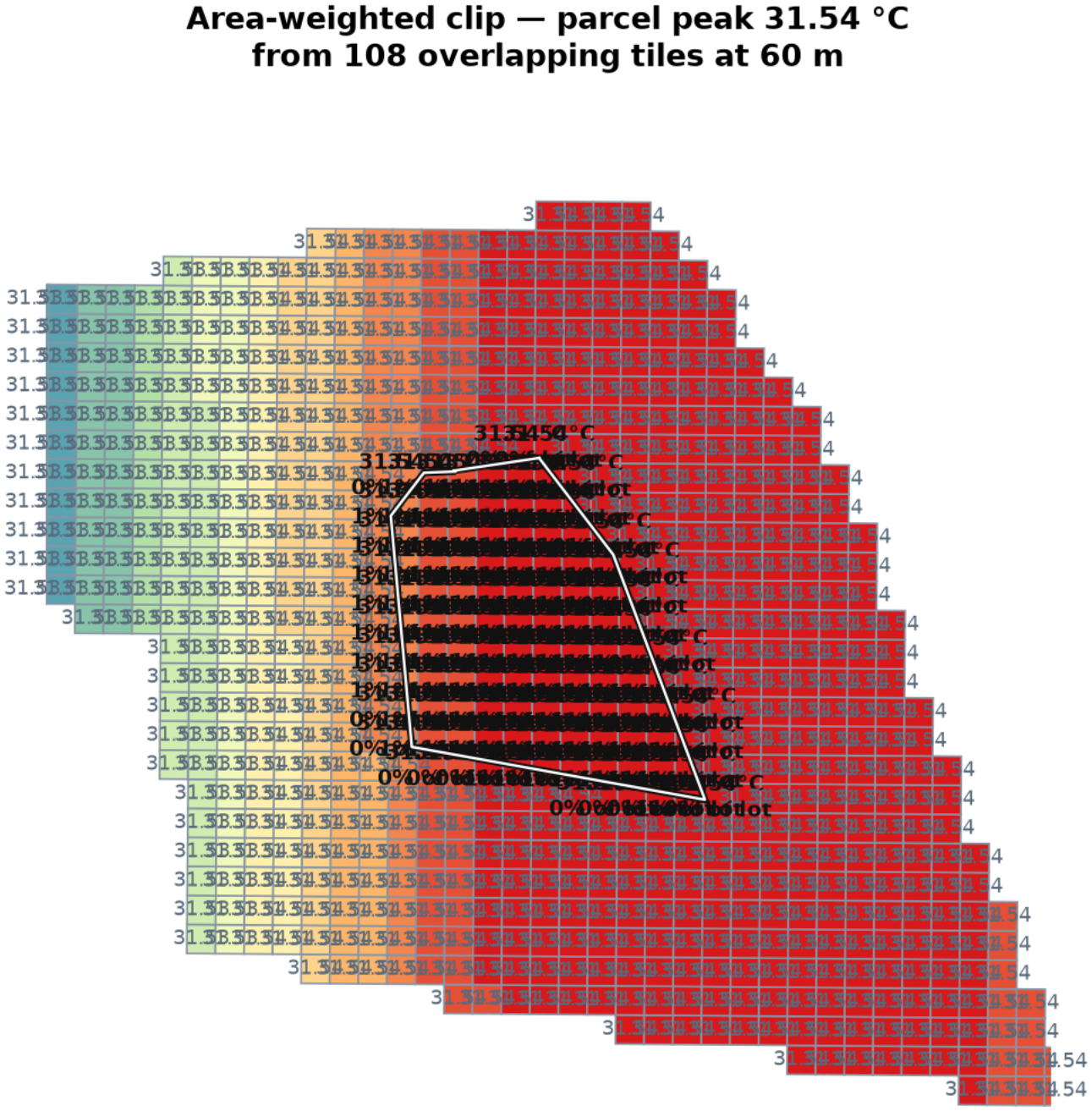
Program: OSHA Heat Illness Prevention

Peak wet-bulb temperature — BELOW

Evaporative and hybrid cooling strategies remain viable on this site.

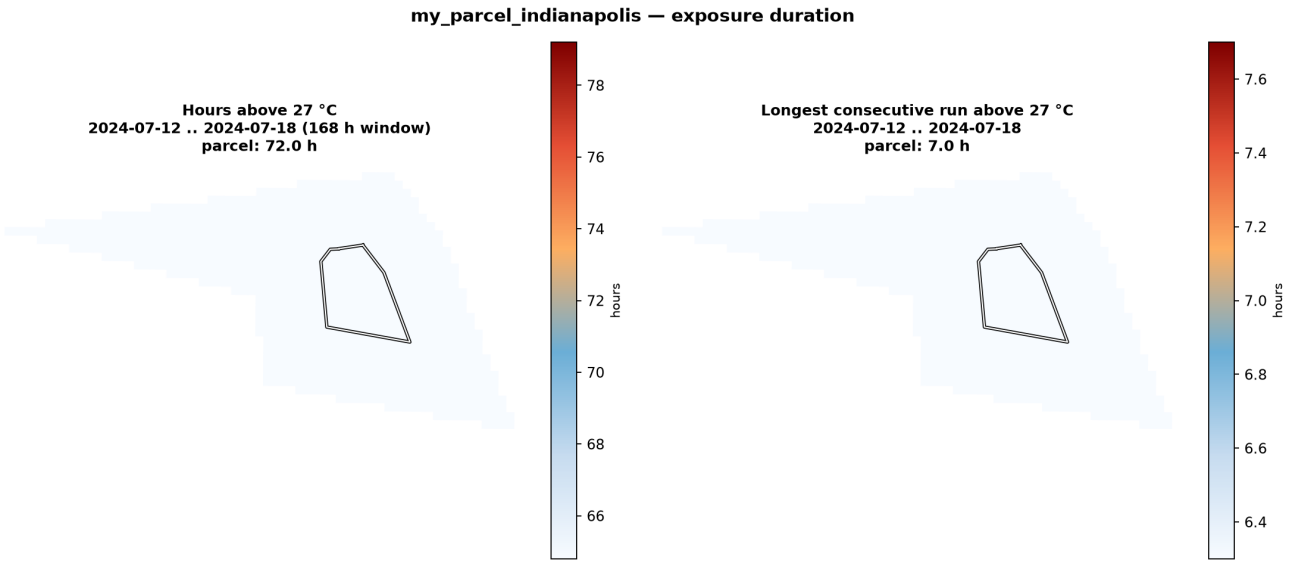
Program: ASHRAE Handbook — HVAC Applications

SITE CONTEXT AND HEAT LAYER



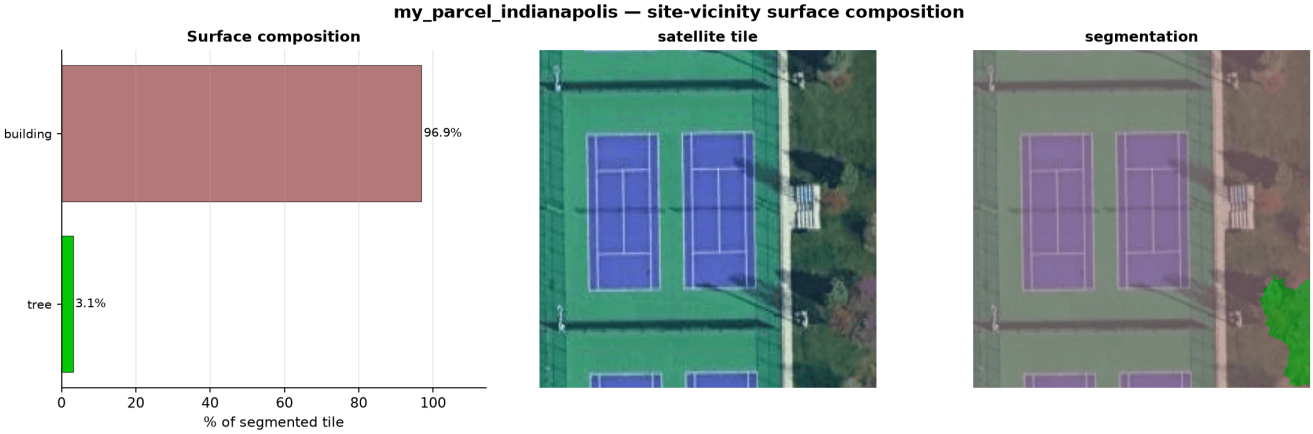
Area-weighted clip: every tile the boundary overlaps, with its share of the lot.

EXPOSURE DURATION



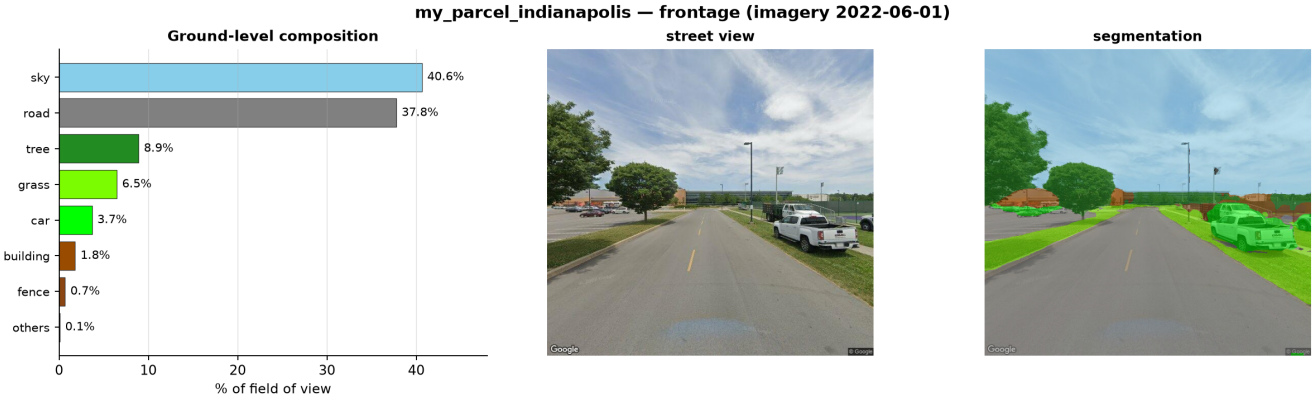
Hours above 27 °C and longest consecutive run, 2024-07-12 to 2024-07-18.

SURFACE COMPOSITION



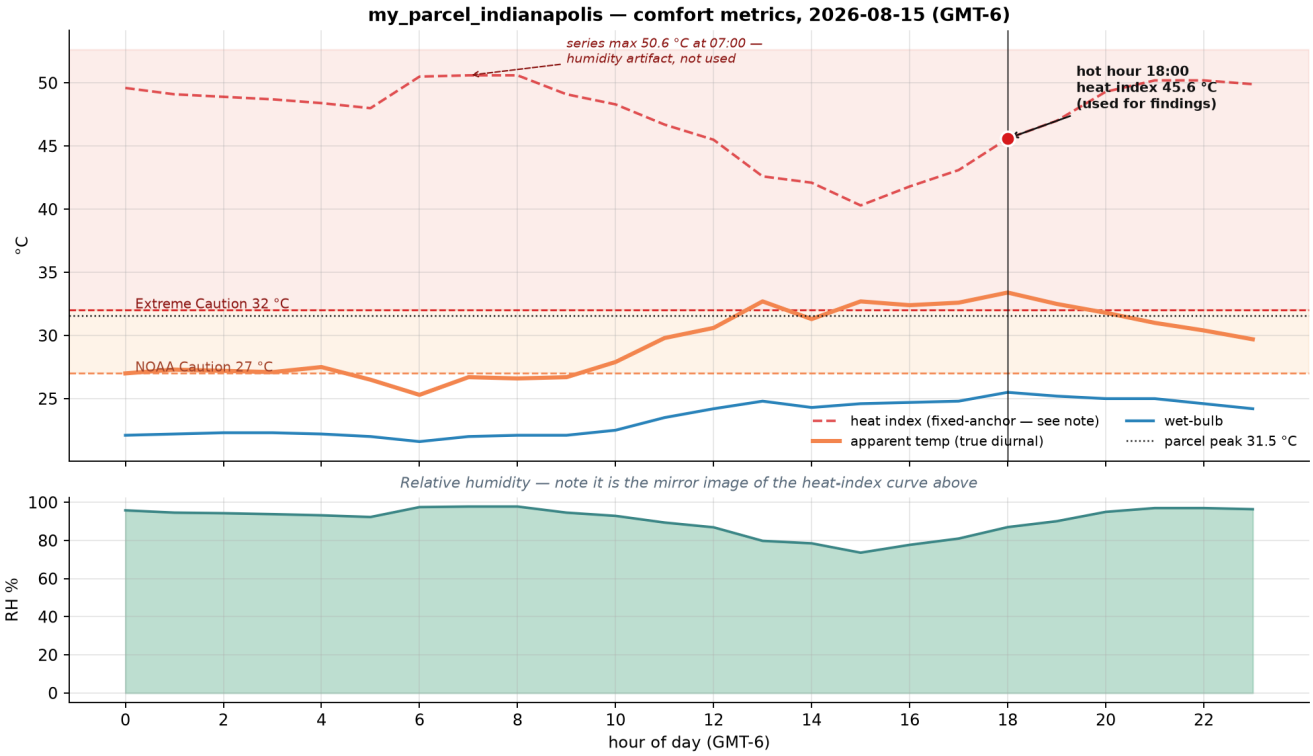
Satellite land-cover segmentation at the site vicinity.

FRONTAGE



Street-view segmentation — the vertical environment at ground level.

COMFORT METRICS



Apparent temperature follows the true diurnal cycle. The heat-index curve (dashed) is derived from a fixed temperature anchor and varies only with humidity, so it inverts — only its value at the hot hour is used.

SOURCES AND LIMITS

Thresholds. NOAA heat-index Caution (27 °C) and Extreme Caution (32 °C); OSHA high-heat trigger (32.2 °C); EPA Heat Island Reduction cool-roof / cool-pavement guidance; USDA Forest Service i-Tree canopy targets; ASHRAE 90.1 and ASHRAE 55.

Data. FortyGuard tOS Enterprise API. Heatmap 2026-08-15 at 60 m; exposure duration 2024-07-12 to 2024-07-18; satellite, street-view, and environmental parameters at the parcel centroid.

Limits. Satellite and street-view segmentation are anchored to the parcel centroid and describe the site vicinity, not the parcel polygon exactly. The heatmap resolves to 60 m, coarser than the parcel, so within-site variation is not resolved. Comfort metrics derive from a single study day and should be re-run against a summer design day before they inform final mechanical sizing.

Heat index. The environmental-parameters endpoint applies one temperature anchor across all 24 hours and varies only humidity, so the heat-index series tracks humidity and peaks overnight — an artifact, not a forecast. This report uses heat index only at the hot hour (18:00, 45.6 °C), where the anchor reflects real conditions. All duration figures come from the measured heatmap exceedance layer, not from this series.